



NAME:

HABIB ID:

LINEAR ALGEBRA

SPRING 2026

SECTIONS L3, L4 & L5

QUIZ 4

Max Marks: 10

Time: 10 minutes

Problem Determine all values of k for which the system has (a) no solution, (b) exactly one solution, (c) infinitely many solutions.

$$\begin{cases} kx + y = 2, \\ x + ky = 2. \end{cases}$$



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Max Marks: 10

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Problem Determine all values of a for which the system has (a) no solution, (b) exactly one solution, (c) infinitely many solutions.

$$\begin{cases} ax + y = 3, \\ x + ay = 3. \end{cases}$$

LINEAR ALGEBRA

Spring 2026

Quiz 4 — Solutions and Marking Scheme

Problem 1 (10 marks)

$$\begin{cases} kx + y = 2, \\ x + ky = 2. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} k & 1 \\ 1 & k \end{pmatrix}.$$

Step 1 (Modeling / determinant)

[4 marks]

$$\det(A) = k^2 - 1 = (k - 1)(k + 1).$$

Step 2 (Exactly one solution)

[2 marks]

$$\det(A) \neq 0 \Rightarrow k^2 - 1 \neq 0 \Rightarrow k \neq \pm 1.$$

Step 3 (Infinite solutions)

[2 marks]

For $k = 1$,

$$x + y = 2, \quad x + y = 2,$$

which are identical. Hence infinitely many solutions.

Step 4 (No solution)

[2 marks]

For $k = -1$,

$$-x + y = 2, \quad x - y = 2,$$

which is inconsistent. Hence no solution.

No solution: $k = -1$,
Exactly one solution: $k \neq \pm 1$,
Infinitely many solutions: $k = 1$.

Problem 2 (10 marks)

$$\begin{cases} ax + y = 3, \\ x + ay = 3. \end{cases}$$

The coefficient matrix is

$$A = \begin{pmatrix} a & 1 \\ 1 & a \end{pmatrix}.$$

Step 1 (Modeling / determinant)

[4 marks]

$$\det(A) = a^2 - 1 = (a - 1)(a + 1).$$

Step 2 (Exactly one solution)

[2 marks]

$$a^2 - 1 \neq 0 \Rightarrow a \neq \pm 1.$$

Step 3 (Infinite solutions)

[2 marks]

For $a = 1$,

$$x + y = 3, \quad x + y = 3,$$

so infinitely many solutions.

Step 4 (No solution)

[2 marks]

For $a = -1$,

$$-x + y = 3, \quad x - y = 3,$$

which is inconsistent.

No solution: $a = -1$,
Exactly one solution: $a \neq \pm 1$,
Infinitely many solutions: $a = 1$.

Grading Note (Method–Independent)

Students may use determinants, row reduction, slope–intercept form, or any other valid method. Marks must be awarded based on correct mathematical reasoning, not on the chosen technique.

For each problem, grading is based on the geometric meaning of the two equations as lines in the plane:

Conceptual step	Marks
Correctly identify when the two equations have the same slope or different slopes (e.g. parallel vs. intersecting lines, or by determinant)	4
Correctly identify the case of two lines intersecting (unique solution)	2
Correctly identify the case of two identical lines (infinitely many solutions)	2
Correctly identify the case of two distinct parallel lines (no solution)	2
Total per question	10

Interpretation for slope–intercept solutions:

- Different slopes $\Rightarrow$ lines intersect $\Rightarrow$ exactly one solution.
- Same slope and same intercept $\Rightarrow$ same line $\Rightarrow$ infinitely many solutions.
- Same slope and different intercepts $\Rightarrow$ parallel lines $\Rightarrow$ no solution.



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SECTIONS L1 & L2

QUIZ 4

Max Marks: 10

Time: 10 minutes

Q. Given A is an $n \times n$ matrix, prove that if the reduced row-echelon form of A is I_n , then A is expressible as a product of elementary matrices.

(Make sure to give full justification and explanation for all steps in your proof).



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QUIZ 4 B SOLUTION & MARK SCHEME

[3 marks]

Assume that the reduced row-echelon form of A is I_n , so that A can be reduced to I_n by a finite sequence of elementary row operations. By Theorem 1.5.1, each of these operations can be accomplished by multiplying on the left by an appropriate elementary matrix. Thus we can find elementary matrices $E_1, E_2, \dots, E_k$ such that

[2 + 2 marks]

$$E_k \cdots E_2 E_1 A = I_n \quad (3)$$

By Theorem 1.5.2, $E_1, E_2, \dots, E_k$ are invertible. Multiplying both sides of Equation 3 on the left successively by $E_k^{-1}, \dots, E_2^{-1}, E_1^{-1}$ we obtain

[3 marks]

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} I_n = E_1^{-1} E_2^{-1} \cdots E_k^{-1} \quad (4)$$

By Theorem 1.5.2, this equation expresses A as a product of elementary matrices.

(since every elementary matrix is invertible, and the inverse is also an elementary matrix).